

1. Use Cases & How It Works

Main Goal:

The AI answers phone calls 24/7, talks like a human receptionist, books appointments, qualifies leads, answers FAQs, and transfers complex calls to humans.

Common Use Cases:

Use Case	Who Uses It	What Happens
Appointment Booking	Clinics, Dentists, Salons	Caller books a slot automatically
Lead Qualification	Real Estate, Agencies	AI asks questions and scores the lead
After-Hours Support	Any business	AI answers FAQs and takes messages
Missed Call Recovery	Service businesses	AI calls back or books callback
FAQ & Information	All businesses	Answers common questions
Call Transfer	All businesses	Transfers to human when needed

2. How the System Works (Step-by-Step)

1. Customer dials your business phone number
 2. Call reaches **Vapi.ai**
 3. Vapi connects to your AI Assistant (with system prompt + tools)
 4. AI talks to the customer in real-time
 5. When needed, AI calls your **Node.js backend** (function calling)
 6. Backend talks to **MongoDB** and **Google Calendar**
 7. Result is sent back to AI → AI continues the conversation
 8. After call ends, full transcript + summary is saved
 9. Web Dashboard and Mobile App show the updated data in real-time
-

3. Detailed Conversation Examples

Example 1: Successful Appointment Booking (Clinic)

AI: Hello! Thank you for calling Smile Dental Clinic. This is Sarah, your virtual assistant. How can I help you today?

Caller: Hi, I want to book a teeth cleaning.

AI: Sure, I'd be happy to help you book a teeth cleaning. May I have your full name, please?

Caller: My name is Ali Khan.

AI: Thank you, Ali. And what's the best phone number to reach you?

Caller: 0300-1234567

AI: Got it. What date and time would you prefer for the cleaning?

Caller: This Friday at 3 PM.

AI: Let me check availability for this Friday at 3 PM...

(AI calls backend → checks Google Calendar)

Yes, 3 PM this Friday is available. Shall I book it for you?

Caller: Yes, please.

AI: Perfect! I've booked your teeth cleaning on Friday at 3 PM. You will receive a confirmation message shortly. Is there anything else I can help you with?

Caller: No, that's all.

AI: Great. We look forward to seeing you, Ali. Have a wonderful day!

Example 2: Lead Qualification (Real Estate)

AI: Hello! Thank you for calling Dream Homes Realty. This is Alex. How may I assist you?

Caller: I'm interested in buying a house in DHA.

AI: Great! I'd love to help. What's your approximate budget?

Caller: Around 3 to 4 crore.

AI: Understood. And when are you looking to buy? Within the next 1 month, 3 months, or later?

Caller: Within 2 months.

AI: Thank you. Are you already pre-approved for a loan, or will you be paying cash?

Caller: I'll need a loan.

AI: Got it. I've noted your requirements. One of our senior agents will contact you within the next 2 hours. Meanwhile, would you like me to send you some available properties on WhatsApp?

Example 3: After Hours / FAQ

AI: Hello! You've reached FitLife Gym. We're currently closed, but I'm here to help. How can I assist you?

Caller: What are your membership prices?

AI: Our monthly membership starts at 8,000 PKR and annual membership is 75,000 PKR. Would you like me to book a free trial session for you?

Example 4: Call Transfer

AI: I understand this is a bit complex. Let me connect you with one of our specialists who can better assist you. Please hold for a moment.

(Call is transferred to human agent with full context)

4. How Appointment Booking Works Internally (With Calendar)

Here's the complete internal flow:

text

1. Caller says they want to book an appointment
2. AI collects: Name, Phone, Date, Time, Reason
3. AI triggers function call → bookAppointment or checkAvailability
4. Request goes to your Node.js Backend
5. Backend does the following:
 - Checks Google Calendar for that date & time
 - If available → Creates event in Google Calendar
 - Saves appointment in MongoDB
 - Optionally sends confirmation SMS/Email/WhatsApp
6. Backend returns success/failure to Vapi
7. AI speaks the result to the caller

Technical Steps in Backend:

1. Receive function call from Vapi
2. Validate data (name, phone, date, time)
3. Convert date & time to proper format
4. Call Google Calendar API:
 - o Check free/busy
 - o Create event if free
5. Save record in MongoDB (appointments collection)
6. Return response to Vapi:
 - o Success → “Appointment booked”
 - o Failure → “That slot is taken, available times are...”

5. How Web Dashboard & Mobile App Will Show the Data

Both Web and Mobile App read data from the **same MongoDB** through your Node.js APIs.

Web Dashboard (React)

Screen	What User Sees
Dashboard	Total calls today, appointments booked, conversion rate, charts
Calls Page	List of all calls + search/filter + transcript + recording

Appointments Page	Calendar view + List view of all bookings
Leads Page	Hot/Warm/Cold leads with scores
Call Detail	Full conversation transcript + AI summary + recording player

Mobile App (React Native)

Screen	What User Sees
Home	Today's summary + recent appointments
Calls	Recent calls list (can tap to see transcript)
Appointments	Upcoming appointments + ability to confirm/cancel
Leads	Hot leads with swipe actions
Notifications	Push notification when new appointment is booked or hot lead comes

Real-time Updates:

- When a new call ends or appointment is booked, both Web and Mobile can refresh data.
- Mobile can also receive **Push Notifications** instantly.